

Sprint 5 – Building Business Logic with Apex

Placement Management System Submission Implemented Features

- Retrieve Student and Job using SOQL
- Prevent duplicate applications
- Validate CGPA, Branch, Backlogs, and Closing Date
- Insert Application records using DML
- Update Application Status
- Comprehensive Apex Test Class with passing tests and 94% coverage

Attached Screenshots

The following screenshots are included in this version of the report: ApplicationService class, ApplicationServiceTest class, Apex Test Execution, and Code Coverage. Additional screenshots (Student, Job, Application records, Execute Anonymous output) can be inserted on the final pages before submission.

```
1 public with sharing class ApplicationService {
2
3     public static String submitApplication(Id studentId, Id jobId) {
4         Student__c student = [
5             SELECT Id, Name, CGPA__c, Branch__c, Active_Backlogs__c, Graduation_Year__c
6             FROM Student__c
7             WHERE Id = :studentId
8             LIMIT 1
9         ];
10        Job__c job = [
11            SELECT Id, Name, Minimum_CGPA__c, Eligible_Branch__c,
12            Backlog_Limit__c, Closing_Date__c
13            FROM Job__c
14            WHERE Id = :jobId
15            LIMIT 1
16        ];
17
18        List<Application__c> existingApplications = [
19            SELECT Id
20            FROM Application__c
21            WHERE Student__c = :studentId
22            AND Job__c = :jobId
23        ];
24
25        if (existingApplications.isEmpty()) {
26            return 'Duplicate Application! Student already applied.';
27        }
28
29        if (student.CGPA__c < job.Minimum_CGPA__c) {
30            return 'Application Rejected: Minimum CGPA not satisfied.';
31        }
32
33        if (student.Branch__c != job.Eligible_Branch__c) {
34            return 'Application Rejected: Branch not eligible.';
35        }
36
37        if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
38            return 'Application Rejected: Too many active backlogs.';
39        }
40
41        if (Date.today() > job.Closing_Date__c) {
42            return 'Application Rejected: Application deadline is over.';
43        }
44
45        Application__c application = new Application__c();
46
47        application.Student__c = studentId;
48        application.Job__c = jobId;
49        application.Application_Date__c = Date.today();
50    }
```

```

33 if (student.Branch__c != job.Eligible_Branch__c) {
34     return 'Application Rejected: Branch not eligible.';
35 }
36
37 if (student.Active_Backlogs__c > job.Backlog_Limit__c) {
38     return 'Application Rejected: Too many active backlogs.';
39 }
40
41 if (Date.today() > job.Closing_Date__c) {
42     return 'Application Rejected: Application deadline is over.';
43 }
44
45 Application__c application = new Application__c();
46
47 application.Student__c = studentId;
48 application.Job__c = jobId;
49 application.Application_Date__c = Date.today();
50 application.Status__c = 'Applied';
51
52 try {
53     insert application;
54
55     return 'Application Submitted Successfully';
56 } catch (Exception e) {
57     return e.getMessage();
58 }
59
60 }
61
62 }
63
64
65 public static String updateApplicationStatus(Id applicationId, String status) {
66
67     Application__c app = [
68         SELECT W.Status__c
69         FROM Application__c
70         WHERE Id = :applicationId
71         LIMIT 1
72     ];
73
74     app.Status__c = status;
75
76     update app;
77
78     return 'Application Status Updated Successfully';
79 }
80
81 }

```



SETUP

Apex Classes

```

1  @isTest
2  public class ApplicationServiceTest {
3
4      @isTest
5      static void testSubmitApplication() {
6
7          // Create Student
8          Student__c student = new Student__c(
9              Name = 'Rahul',
10             Email__c = 'rahul@gmail.com',
11             CGPA__c = 8.5,
12             Branch__c = 'CSE',
13             Active_Backlogs__c = 0,
14             Graduation_Year__c = 2027
15         );
16         insert student;
17
18         // Create Job
19         Job__c job = new Job__c(
20             Name = 'Software Engineer',
21             Company__c = 'TCS',
22             Minimum_CGPA__c = 7.5,
23             Eligible_Branch__c = 'CSE',
24             Backlog_Limit__c = 0,
25             Closing_Date__c = Date.today().addDays(10)
26         );
27         insert job;
28
29         Test.startTest();
30
31         String result =
32             ApplicationService.submitApplication(student.Id, job.Id);
33
34         Test.stopTest();
35
36         System.assertEquals(
37             'Application Submitted Successfully',
38             result
39         );
40     }
41
42     @isTest
43     static void testDuplicateApplication() {
44
45         Student__c student = new Student__c(
46             Name = 'Student1',
47             Email__c = 'student1@gmail.com',
48             CGPA__c = 8,
49             Branch__c = 'CSE',
50             Active_Backlogs__c = 0,
51             Graduation_Year__c = 2027
52         );
53         insert student;
54
55         Job__c job = new Job__c(
56             Name = 'Developer',
57             Company__c = 'Infosys',
58             Minimum_CGPA__c = 7,
59             Eligible_Branch__c = 'CSE',
60             Backlog_Limit__c = 0,
61             Closing_Date__c = Date.today().addDays(5)
62         );
63         insert job;
64
65         ApplicationService.submitApplication(student.Id, job.Id);
66
67         String result =

```

```

67 string result =
68     ApplicationService.submitApplication(student.Id, job.Id);
69
70     System.assert(
71         result.contains("Duplicate")
72     );
73 }
74 @isTest
75 static void testLowCGPA() {
76
77     Student__c student = new Student__c(
78         Name="Low CGPA",
79         Email__c="low@test.com",
80         CGPA__c=6.0,
81         Branch__c="CSE",
82         Active_Backlogs__c=0,
83         Graduation_Year__c=2027
84     );
85     insert student;
86
87     Job__c job = new Job__c(
88         Name="Developer",
89         Company__c="Infosys",
90         Minimum_CGPA__c=7.5,
91         Eligible_Branch__c="CSE",
92         Backlog_Limit__c=0,
93         Closing_Date__c=Date.today().addDays(5)
94     );
95     insert job;
96
97     String result =
98         ApplicationService.submitApplication(student.Id, job.Id);
99
100     System.assert(result.contains("CGPA"));
101 }
102 @isTest
103 static void testBranchValidation() {
104
105     Student__c student = new Student__c(
106         Name="Branch Test",
107         Email__c="branch@test.com",
108         CGPA__c=9,
109         Branch__c="ECE",
110         Active_Backlogs__c=0,
111         Graduation_Year__c=2027
112     );
113     insert student;
114
115     Job__c job = new Job__c(
116         Name="Developer",
117         Company__c="Infosys",
118         Minimum_CGPA__c=7,
119         Eligible_Branch__c="CSE",
120         Backlog_Limit__c=0,
121         Closing_Date__c=Date.today().addDays(5)
122     );
123     insert job;
124
125     String result =
126         ApplicationService.submitApplication(student.Id, job.Id);
127
128     System.assert(result.contains("Branch"));
129 }
130 @isTest
131 static void testBacklogValidation() {
132
133     Student__c student = new Student__c(
134         Name="Backlog Test",

```

Apex Classes

```

145 // Job__c job = new Job__c(
146 //     Name="Developer",
147 //     Company__c="Infosys",
148 //     Minimum_CGPA__c=7,
149 //     Eligible_Branch__c="CSE",
150 //     Backlog_Limit__c=0,
151 //     Closing_Date__c=Date.today().addDays(5)
152 // );
153 // insert job;
154
155 String result =
156     ApplicationService.submitApplication(student.Id, job.Id);
157
158 System.assert(result.contains("Branch"));
159 }
160 @isTest
161 static void testBranchValidation() {
162
163     Student__c student = new Student__c(
164         Name="Branch Test",
165         Email__c="branch@test.com",
166         CGPA__c=9,
167         Branch__c="ECE",
168         Active_Backlogs__c=0,
169         Graduation_Year__c=2027
170     );
171     insert student;
172
173     Job__c job = new Job__c(
174         Name="Developer",
175         Company__c="Infosys",
176         Minimum_CGPA__c=7,
177         Eligible_Branch__c="CSE",
178         Backlog_Limit__c=0,
179         Closing_Date__c=Date.today().addDays(5)
180     );
181     insert job;
182
183     String result =
184         ApplicationService.submitApplication(student.Id, job.Id);
185
186     System.assert(result.contains("Branch"));
187 }
188 @isTest
189 static void testBacklogValidation() {
190
191     Student__c student = new Student__c(
192         Name="Backlog Test",

```

Application Test Execution

To run one or more Apex or flow unit tests, click Select Tests. To view the current Apex code coverage for an individual class or your organization, go to the [Apex Classes](#) page.



In Winter '26 and later, this page shows test details for multiple categories of tests, including Apex and flow tests. For more info about how to [Test Your Changes](#), go to Salesforce Help.

Select Tests... Developer Console Options... [View Test History](#)

Abort			
☐	Status	Class	Result
Test Run: 2026-08-03 11:59:46, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☐		[View] ApplicationServiceTest	
Test Run: 2026-08-03 11:34:33, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(7/7) Test Methods Passed
Test Run: 2026-08-03 11:34:26, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(7/7) Test Methods Passed
Test Run: 2026-08-03 11:30:10, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(2/2) Test Methods Passed
Test Run: 2026-08-03 11:29:53, jayasri73737@resourceful-narwhal-rffxm7.com, (1 test class run)			
☒		[View] ApplicationServiceTest	(2/2) Test Methods Passed

Code coverage



Apex Class
ApplicationService

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Apex Class Detail

[Edit](#) [Delete](#) [Download](#) [Security](#) [Show Dependencies](#)

Name	ApplicationService	Status	Active
Namespace Prefix		Code Coverage	94% (32/34)
Created By	Java Sri Padilam, 03/08/2026, 10:33 am	Last Modified By	Java Sri Padilam, 03/08/2026, 11:54 am

[Class Body](#) [Class Summary](#) [Version Settings](#) [Trace Flags](#)

Application Record

Student

Swathi

New ContactEditNew Opportunity

Related

Details

Student Name

Swathi

Owner

Jaya Sri Padilam

Email

swathi@gmail.com

CGPA

8.50

Branch

CSE

Active Backlogs

0

Graduation Year

2027

Created By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Last Modified By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Job record

Job

Software Engineer

New ContactEditNew Opportunity

Related

Details

Job Title

Software Engineer

Owner

Jaya Sri Padilam

Company

TCS

Minimum CGPA

7.50

Closing Date

31/06/2026

Eligible Branch

CSE

Backlog Limit

0

Created By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Last Modified By

Jaya Sri Padilam, 03/06/2026, 10:31 am

Execute Anonymous output

The screenshot shows the Salesforce Developer Console interface. The top bar indicates the browser is Google Chrome, and the address bar shows the URL: `resourceful-narwhal-rffxm7-dev-ed.trailblaze.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage`. The tabs include `9/07/2026, 10:25:00`, `AppointmentFuture.apxc`, `Log executeAnonymous @8/3/2026, 11:40:21 AM`, and `Log executeAnonymous @8/3/2026, 11:45:05 AM`.

The **Execution Log** section displays a table with the following data:

Timestamp	Event	Details
11:45:05:059	USER_DEBUG	[4]DEBUG Duplicate Application! Student already applied.

Below the log, there is a filter section with checkboxes for `This Frame`, `Executable`, `Debug Only` (checked), and `Filter`. A text input field says "Click here to filter the log".

The bottom section shows a table with the following data:

User	Application	Operation	Time	Status	Read	Size
Jaya Sri Padilam	Unknown	/services/data/v67.0/t...	8/3/2026, 11:45:05 AM	Success		6.87 KB

Github Repository

<https://github.com/7jayasri/Placement-Management-System>